Seasonal Variations and Long-Term Trends in Nutrient and Sediment Concentrations in the Lower Mississippi River

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Abstract

Nutrient inputs from the Mississippi River contribute to a seasonal low-oxygen “dead zone” in the northern Gulf of Mexico, affecting marine habitats and organisms. Precipitation across the basin can influence runoff, river discharge, and the transport of nutrients and sediment. This study will analyze USGS water-quality data from the Mississippi River near St. Francisville, Louisiana, for water years 2008–2025, focusing on nitrate, total phosphorus, and suspended-sediment concentrations. Using Python, seasonal differences will be examined with boxplots, while long-term trends will be assessed with the Mann–Kendall test and Sen’s slope estimator. Spearman rank correlation will be used to examine relationships among basin-wide precipitation, river discharge, and constituent concentrations. The findings will help clarify how water quality varies over time and relates to precipitation and discharge, providing useful information for water-quality monitoring and nutrient management.

1. Introduction

Each summer, a large area of bottom water off the Louisiana coast becomes low in oxygen. This "dead zone," or hypoxia, happens when excess nitrogen and phosphorus from the Mississippi River feed algae, and bacteria use up the oxygen while breaking down the dead algae (Rabalais et al., 2002). Hypoxia affects the Gulf’s ecology by reducing suitable habitat for fish and shrimp, including Atlantic croaker and brown shrimp (Craig & Crowder, 2005). These effects also impact fisheries and the livelihoods of people in Louisiana who depend on them (Rabalais & Turner, 2019).

The Mississippi River receives nutrients from a large watershed that drains about 41% of the contiguous United States and includes extensive farmland (Figure 1). Nitrate delivery to the Gulf nearly tripled between 1955–1970 and 1980–1999, driven by increased fertilizer use combined with hydrologic factors such as precipitation and streamflow (Goolsby & Battaglin, 2001). Meanwhile, sediment delivery has declined, partly because dams trap sediment before it reaches the lower river (Meade & Moody, 2010). This decline is a concern for coastal wetlands that depend on river sediment.

Rainfall carries nutrients and soil from the land into rivers. Wet winters and springs in agricultural areas can increase nitrogen delivery to the Gulf (Donner & Scavia, 2007), and future changes in precipitation may increase nitrogen loading further (Sinha et al., 2017). Nutrient and sediment concentrations can also vary by season. In spring, rain and snowmelt can carry nutrients and soil from fields into streams. During summer, river flow is often lower, and plants and algae take up nitrate. These processes can affect concentration. Over the years, changes in farming practices, land use, and river management can influence water quality (Meade & Moody, 2010).

The Hypoxia Task Force aimed to reduce nitrogen and phosphorus loading by 20% by 2025, using 1980–1996 as the baseline (US EPA, 2015). A USGS study examined changes in river flow and nutrient transport from 1980 to 2007 (Battaglin et al., 2010). This study will examine seasonal variations and long-term trends in nutrient and sediment concentrations in the Mississippi River near St.

Francisville, Louisiana, for water years 2008–2025. It will also explore how these concentrations relate to precipitation and river discharge.

2. Research Questions and Hypotheses

Question 1: Do nitrate, total phosphorus, and suspended sediment concentrations differ among seasons (winter, spring, summer, and fall)?
Hypothesis 1: Concentrations will differ among seasons. Nitrate will be highest in spring and lowest in late summer and fall, because spring rain washes fertilizer from farmland into the river and plants and algae take up more nitrate in summer. Total phosphorus and suspended sediment will be highest in spring, when river flow is high, and more soil is washed into the river.
Question 2: Have nitrate, total phosphorus, and suspended sediment concentrations increased, decreased, or stayed the same between water years 2008 and 2025?
Hypothesis 2: Suspended sediment concentrations will decrease, continuing the long-term decline reported by Mize et al. (2018). The analysis of nitrate and total phosphorus is exploratory.
Question 3: How are monthly concentrations related to basin-wide precipitation and to river discharge?
Hypothesis 3: Nitrate, total phosphorus, and suspended sediment concentrations will be positively related to both discharge and precipitation, because heavy rain washes nutrients and soil from the land into rivers (Donner & Scavia, 2007).

3. Methods

3.1 Study site and data

The USGS station on the Mississippi River near St. Francisville, Louisiana has been selected because it is one of the two stations USGS uses to measure the amount of nutrients and sediment the river delivers to the Gulf (Figure 1). It sits near the end of the river, so it allows the investigation of inputs from the upstream basin. The station also has enough data for seasonal and trend tests. As Battaglin et al. (2010) studied nutrient trends for the study period between 1980 and 2007, the current study period is water years 2008–2025.

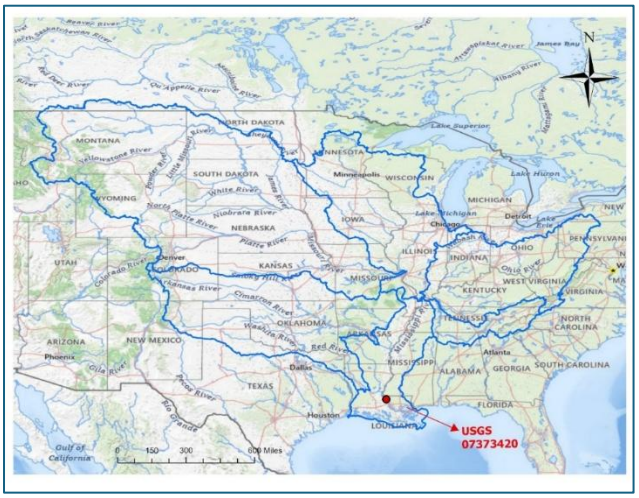


Figure 1. Mississippi-Atchafalaya River Basin with its six major sub-basins and the USGS station Mississippi River near St. Francisville, Louisiana (07373420).

Table 1 lists the data that will be used in this project.

Table 1. Data used in this project.

Data	Source
Nitrate, total phosphorus, and suspended sediment concentrations	USGS 07373420
Monthly river discharge	USGS data release
Monthly precipitation for six sub-basins	NOAA NCEI Climate at a Glance
Sub-basin boundaries and areas	Shapefile of the six Mississippi River sub-basins

As a preliminary result, the median discharge for each month has been plotted. Discharge at St. Francisville follows a strong seasonal cycle (Figure 2). This suggests that nutrient and sediment concentrations may also differ among seasons.

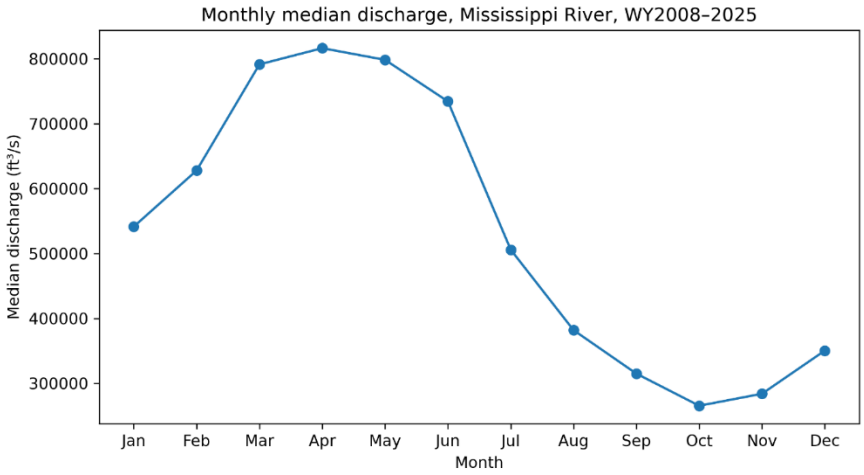


Figure 2. Preliminary result: median monthly discharge of the lower Mississippi River, water years 2008–2025.

3.2 Data Preparation

All data processing will be done in Python, using pandas for data handling, pymannkendall and SciPy for statistics, and matplotlib for figures. The water-quality file from the Water Quality Portal will be read, and only nitrate (USGS parameter code 00631), total phosphorus (00665), and suspended sediment (80154) for water years 2008–2025 will be kept. The station is sampled about once a month. Each monthly value will then be assigned to a season (winter = December–February, spring = March–May, summer = June–August, fall = September–November), and one median will be calculated per water year for the trend test. For precipitation, the six sub-basin files will be combined into one basin-wide value per month using an area-weighted average, so that larger sub-basins count more. Finally, the monthly concentration, discharge, and precipitation values will be joined into one table.

3.3 Question 1: Seasonal differences

For each constituent, a boxplot with one box per season will be made to show when concentrations are highest and lowest.

3.4 Question 2: Long-term trends

The annual median concentration for each water year will be plotted, and a trend will be tested using the Mann–Kendall test (Kendall, 1975; Mann, 1945). This test checks whether values tend to go up or down over time. A p-value below 0.05 will indicate a significant trend. The size of the trend will be measured with Sen's slope (Sen, 1968), the median of the slopes between all pairs of years.

3.5 Question 3: Precipitation, discharge, and concentration

Question 3 will be answered using Spearman rank correlation (Spearman, 1904). The Spearman coefficient ranges from −1 to +1. A positive value means both variables increase together, a negative value means one decreases as the other increases, and a value near 0 means no clear relationship.

1. **Discharge and concentration:** Each constituent will be correlated with monthly discharge. A positive relationship would mean higher flow washes more material into the river; a negative relationship would mean the extra water dilutes it.
2. **Precipitation and concentration:** Each constituent will be correlated with monthly basin-wide precipitation.

Each relationship will be shown with a scatter plot.

4. Timeline

The project will take four weeks, at about 10–12 hours per week. In Weeks 1 and 2, the data will be explored and prepared, including calculating monthly and annual medians, assigning seasons, and building the basin-wide precipitation series (Section 3.2). In Week 3, the data will be analyzed using seasonal boxplots, the Mann–Kendall test and Sen's slope, and Spearman correlation (Sections 3.3–3.5). Finally, in Week 4, the final report will be completed.

5. Broader Impacts

This project will describe nutrient and sediment conditions in the lower Mississippi River from 2008 to 2025, covering the years after the study by Battaglin et al. (2010). Understanding when concentrations are highest and how they relate to rainfall may help agencies plan water-quality monitoring and nutrient-reduction efforts. Improving water quality matters for Gulf fisheries and the Louisiana communities that depend on them. Sediment patterns may also provide useful information for coastal restoration, since river sediment helps build and maintain wetlands. The project will also include a documented Python workflow that can be used again as new USGS data becomes available.

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